
Social Networks and their Analysis

BASIC CONCEPTS

BY: IVETA MRAZOVA



Social Networks and their Analysis

An Introduction to the field

- *Motivation*
 - *Link Analysis*
 - *SF-networks*
- *Background*
 - *Application Examples*

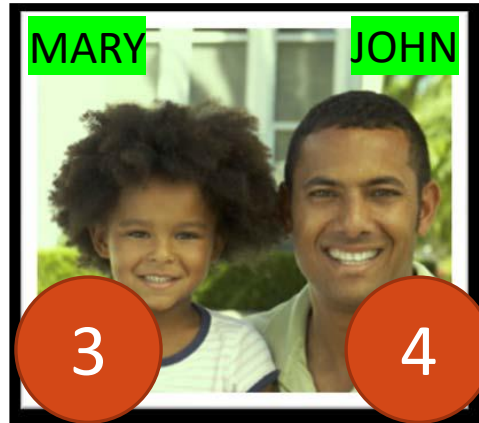
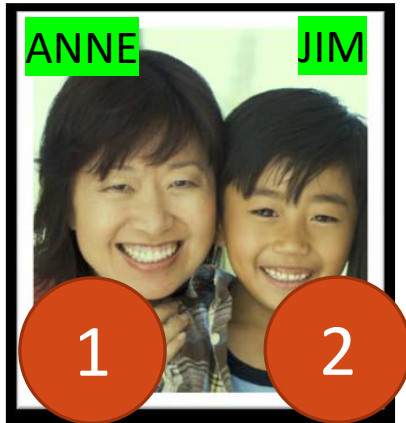
Basic Concepts

- **Network Representation**
- Tie Strength
- Key Players
- Cohesion

Basic Concepts

- **Networks** How to represent various social networks?
- **Tie Strength** How to identify strong / weak ties in the network?
- **Key Players** How to identify key / central nodes in the network?
- **Cohesion** Measures of overall network structure

Representing Relations as Networks



Communication

Anne: Jim, tell the Murrys they're invited.

Jim: Mary, you and your dad should come for dinner!

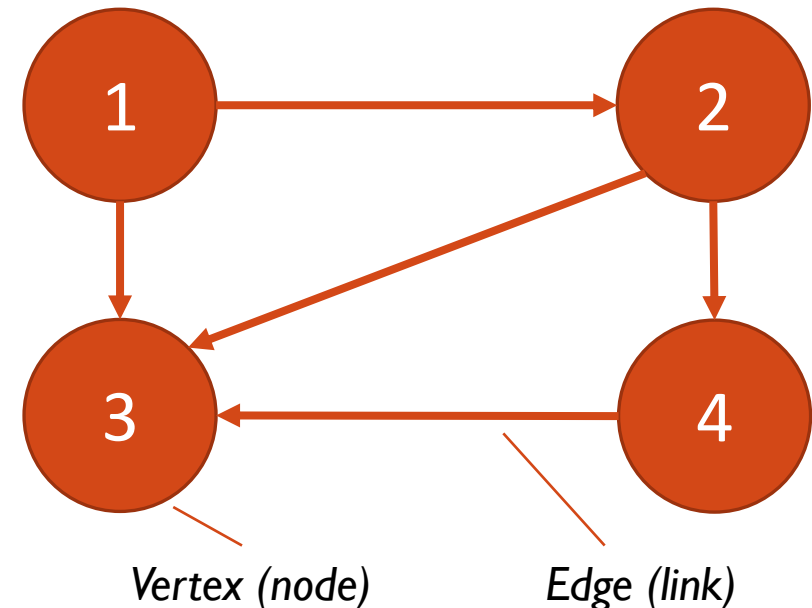
Jim: Mr. Murray, you should both come for dinner

Anne: Mary, did Jim tell you about the dinner? You must come.

John: Mary, we are invited for dinner tonight, it's settled!

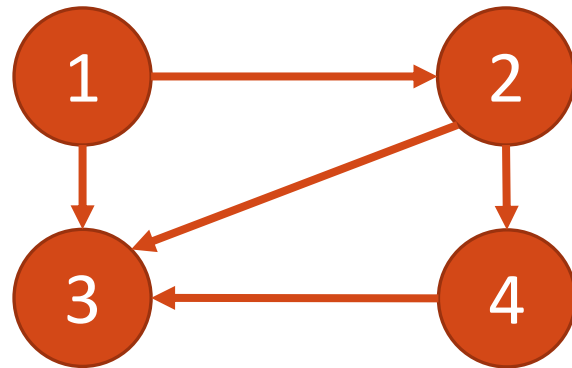
Can we study their interactions as a network?

Graph:



Entering Data on a Directed Graph

A (directed) graph



Edge list

Vertex Vertex

1	2
1	3
2	3
2	4
4	3

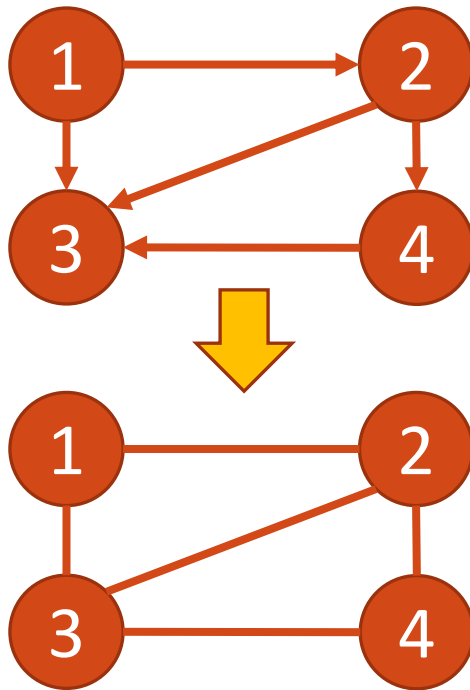
Adjacency matrix

Vertex	1	2	3	4
1	0	1	1	0
2	0	0	1	1
3	0	0	0	0
4	0	0	1	0

For directed graphs, the adjacency matrix does not have to be symmetric.

Representing an Undirected Graph

Directed: who contacts whom



Undirected: who knows whom

Same edge list

X different interpretation

Vertex Vertex

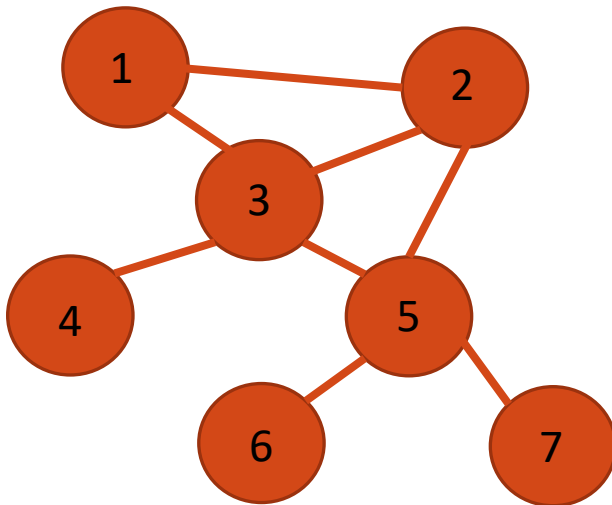
1	2
1	3
2	3
2	4
3	4

Symmetric adjacency matrix

Vertex	1	2	3	4
1	0	1	1	0
2	1	0	1	1
3	1	1	0	1
4	0	1	1	0

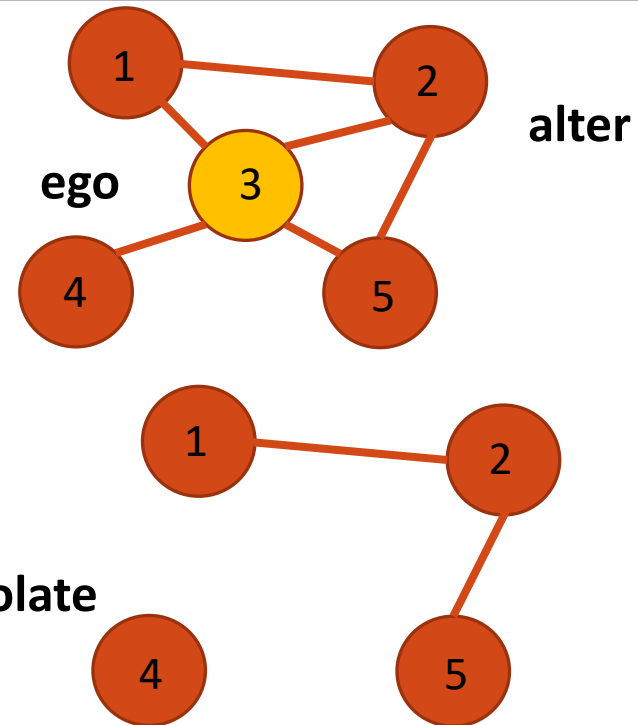
Ego Networks and „Whole“ Networks

a “whole” network*



3's ego network

3's ego network
without ego**



* No studied network is „whole“ in practice: it is usually a partial picture of one's real-life network (boundary specification problem)

** Ego is not needed for further analysis as all alters are by definition connected to ego

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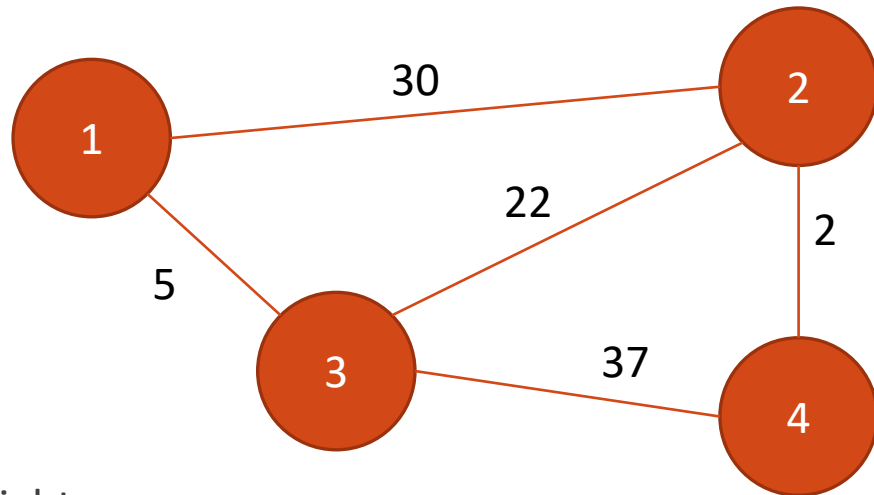
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Adding Weights to (Directed or Undirected) Edges



The weights can express:

- frequency of interaction within the period of observation
- number of items exchanged within a period
- individual perceptions of strength of relationship
- costs in communication or exchange, e.g., distance
- combinations of the above

Edge list: add column of weights

Vertex	Vertex	Weight
1	2	30
1	3	5
2	3	22
2	4	2
3	4	37

Adjacency matrix: add weights

Vertex	1	2	3	4
1	0	30	5	0
2	30	0	22	2
3	5	22	0	37
4	0	2	37	0

Edge Weights as Relationship Strength

- **Edges** can represent interactions, flows of information or goods, similarities/affiliations, or social relations
- Specifically for social relations, a **‘proxy’ for the strength of a tie** can be:
 - the frequency of interaction (communication) or the amount of flow (exchange)
 - reciprocity in interaction or flow
 - the type of interaction or flow between the two parties (e.g., intimate or not)
 - other attributes of the nodes or ties (e.g., kin relationships)
 - the structure of the nodes’ neighborhood (e.g., many mutual ‘friends’)
- Surveys and interviews can reveal the existence of mutual or one-sided strength/affection with greater certainty, but proxies are also useful

Homophily, Transitivity, and Bridging

Homophily ~ tendency to relate to people with similar characteristics (status, beliefs, etc.)

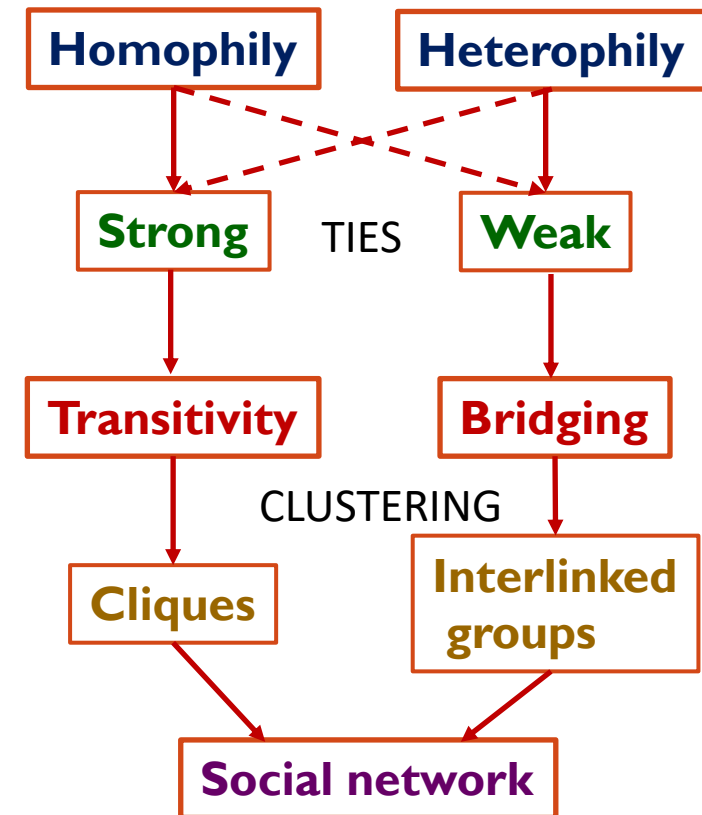
- tendency to homogeneous groups (clusters) where forming relations is easier
- extreme homogenization can act counter to innovation and idea generation
=> **heterophily** is thus desirable in some contexts
- homophilous ties can be strong or weak

Transitivity ~ 'if there is a tie between A and B and a tie between B and C, then A and C will also be connected in a transitive network'

- strong ties are more often transitive than weak ties; transitivity thus indicates the existence of strong ties (but is not a necessary or sufficient condition)
- transitivity and homophily together lead to the formation of cliques (fully connected clusters)

Bridges ~ nodes and edges that connect different groups

- usually weak ties, but not every weak tie is a bridge
- facilitate inter-group communication, increase social cohesion, and help spur innovation



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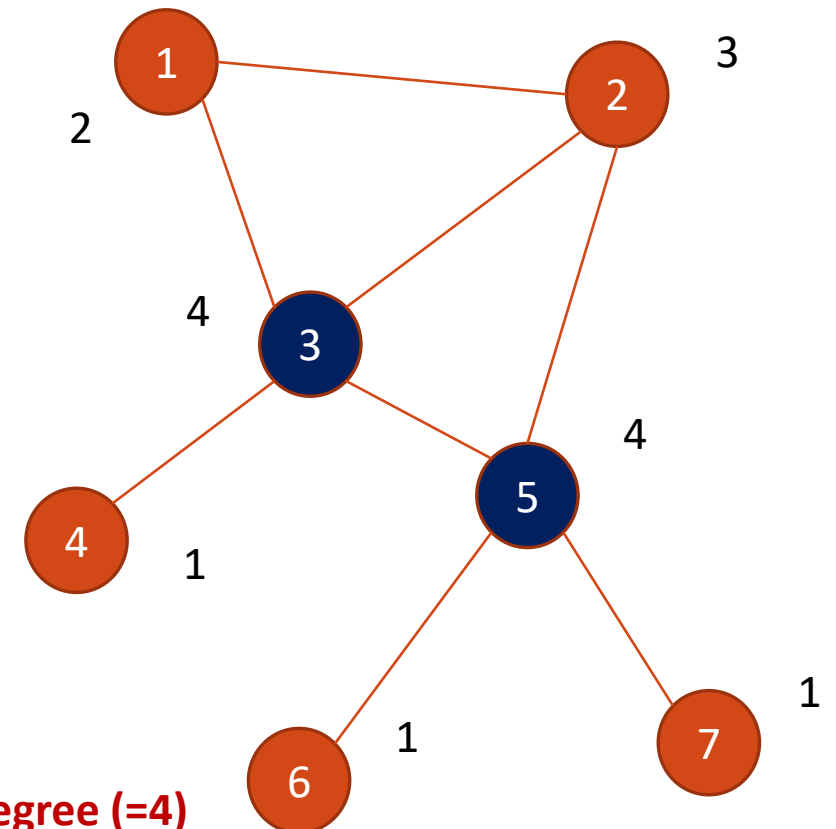
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Degree centrality

node's in- and out- degree:

- ~ number of links that lead into / out of the node
- in undirected graphs, in- and out- degrees are identical
- measures node's connectedness, influence and / or popularity
- helps assess which nodes are central with respect to spreading information and influencing others in their 'neighborhood'

A hypothetical graph

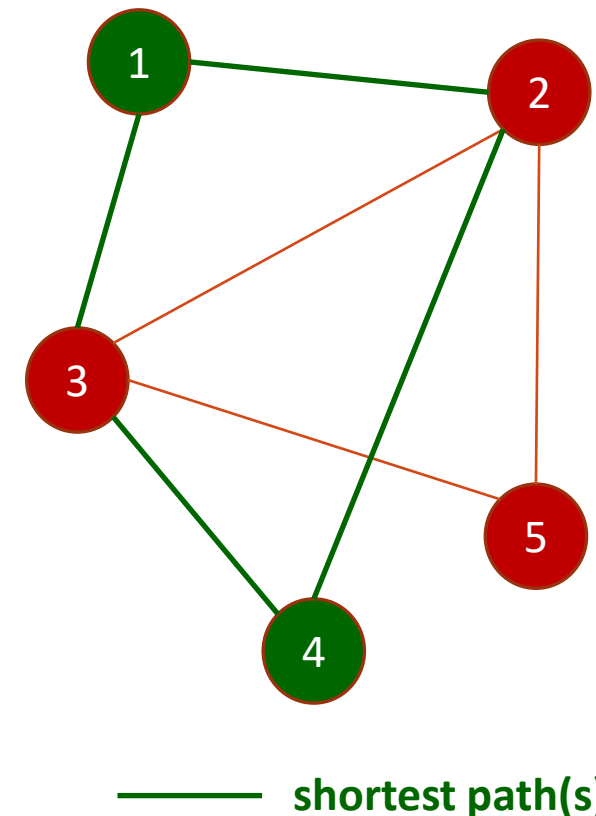


Nodes 3 and 5 have the highest degree (=4)

Paths and shortest paths

- a **path** between two nodes is any sequence of non-repeating nodes that connects the two nodes
- the **shortest path** between two nodes is the path that connects the two nodes with the shortest number of edges (distance between the nodes)
- shorter paths are preferred for quick communication or exchange (but sometimes not, e.g., when spreading disease)
- **Example:**
 - between the nodes 1 and 4, there are two shortest paths of length 2 – {1,2,4} and {1,3,4}
 - other paths between 1 and 4 are {1,2,3,4}, {1,3,2,4}, {1,2,5,3,4} and {1,3,5,2,4} (the longest paths)

A hypothetical graph



Shortest path algorithms

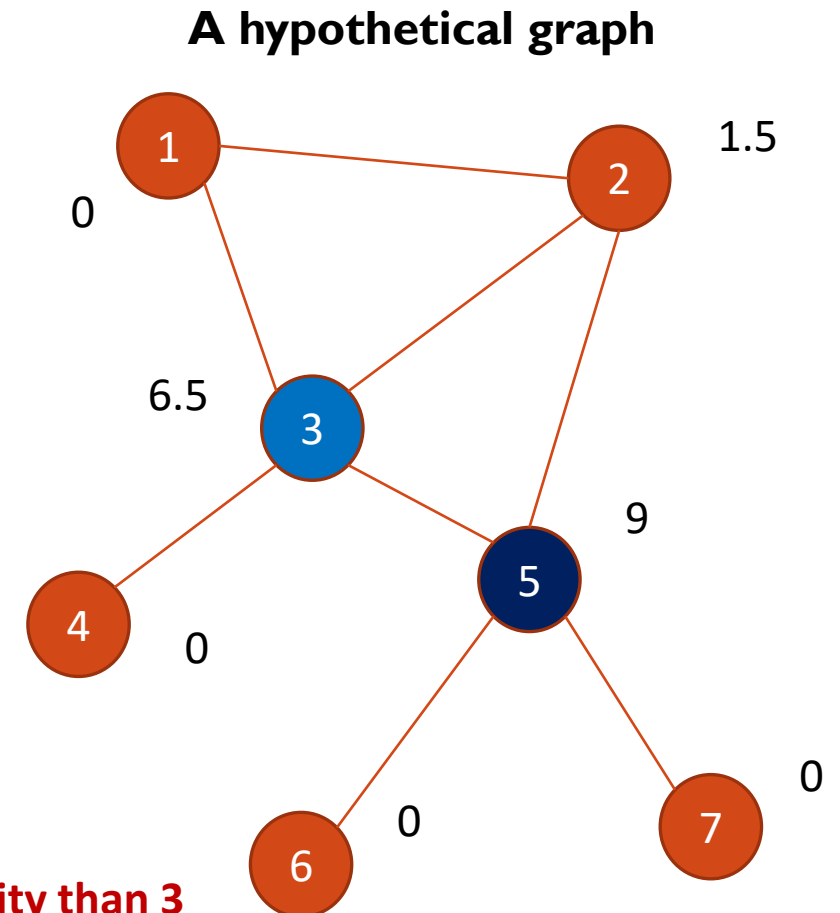
B. V. Cherkassky, A. V. Goldberg, T. Radzik: Shortest paths algorithms: Theory and experimental evaluation, in: Mathematical Programming, 73, 1996, pp. 129-174.

- **Single source:** Given a graph $G=(V,E)$, with weight function $w(u,v) = w_{u,v}$, and a single source vertex, s , return the shortest paths from s to all other vertices in V .
 - General graphs ($\sim$ directed, negative edge weights allowed): **Bellman-Ford algorithm** with the complexity $O(|V| \cdot |E|)$
 - Graphs without negative edge weights: **Dijkstra's algorithm** with the complexity $O(|V|^2)$
 - To find the shortest path between only two vertices, the algorithm can stop after that shortest path is found.
- **All pairs:** Given a graph $G=(V,E)$, with weight function $w(u,v) = w_{u,v}$, return the shortest path from u to v , $\forall u,v \in V$.
 - Graphs with negative edge weights allowed: **Floyd-Warshall algorithm** with the complexity $O(|V|^3)$
 - Sparser graphs with negative edge weights allowed: **Johnson's algorithm** with the complexity $O(|E| \cdot |V| + |V|^2 \cdot \log_2(|V|))$
 - Graphs without negative edge weights: **Dijkstra's algorithm** with the complexity $O(|V|^3)$

Betweenness centrality

- ~ number of shortest paths passing through a node divided by all shortest paths in the network
 - for a given node v , calculate the number of shortest paths between nodes i and j that pass through v , and divide by all shortest paths between nodes i and j
 - sum the computed values for all node pairs i, j
 - sometimes normalized such that the highest value is 1
- indicates nodes that are on communication paths between other nodes
- helps determine points where the network might break apart (think, e.g., of cutting off nodes 3 or 5)

Node 5 has higher betweenness centrality than 3

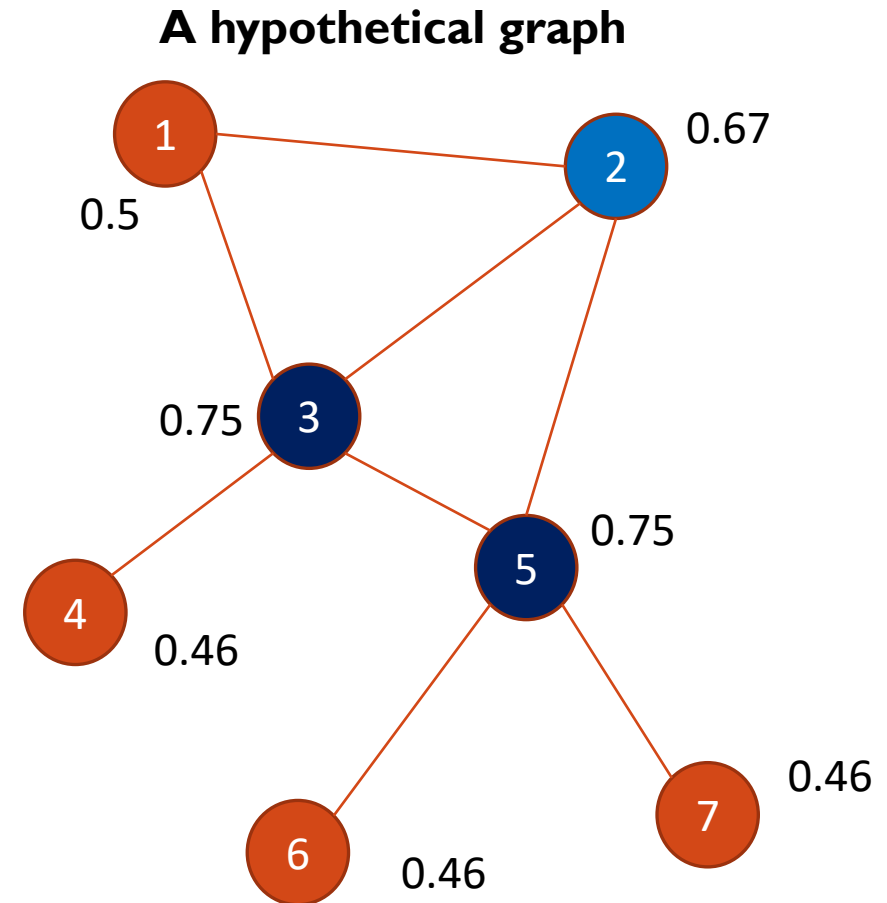


Closeness centrality

Closeness centrality is a measure of reach

- ~ speed with which information can reach other nodes from a given starting node
 - calculate the mean length of all shortest paths from a node to all other nodes in the network
 - ~ how many hops does it take on average to reach every other node
 - take the reciprocal of the above value
- higher values indicate higher closeness (are 'better')
- useful when the speed of dissemination is of main concern

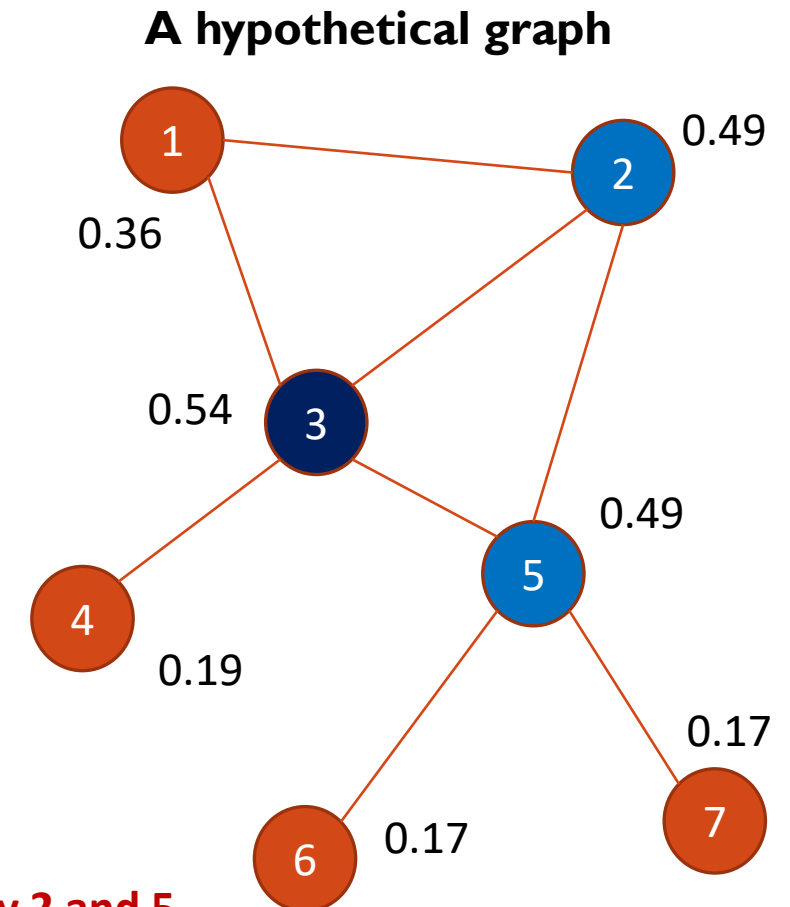
Nodes 3 and 5 have the highest (best) closeness, while node 2 is almost as good



Eigenvector centrality

- ~ proportional to the sum of eigenvector centralities of all nodes directly connected to the considered node
 - a node with a high eigenvector centrality is connected to other nodes with a high eigenvector centrality
 - similar to Google's PageRank, used to rank web pages
 - links from highly linked-to pages count more
 - useful in determining who is connected to the most connected nodes
 - actually, measured centrality values may slightly vary based on the software used

Node 3 has the highest eigenvector centrality, closely followed by 2 and 5



Interpretation of measures

Centrality measure Interpretation in social networks

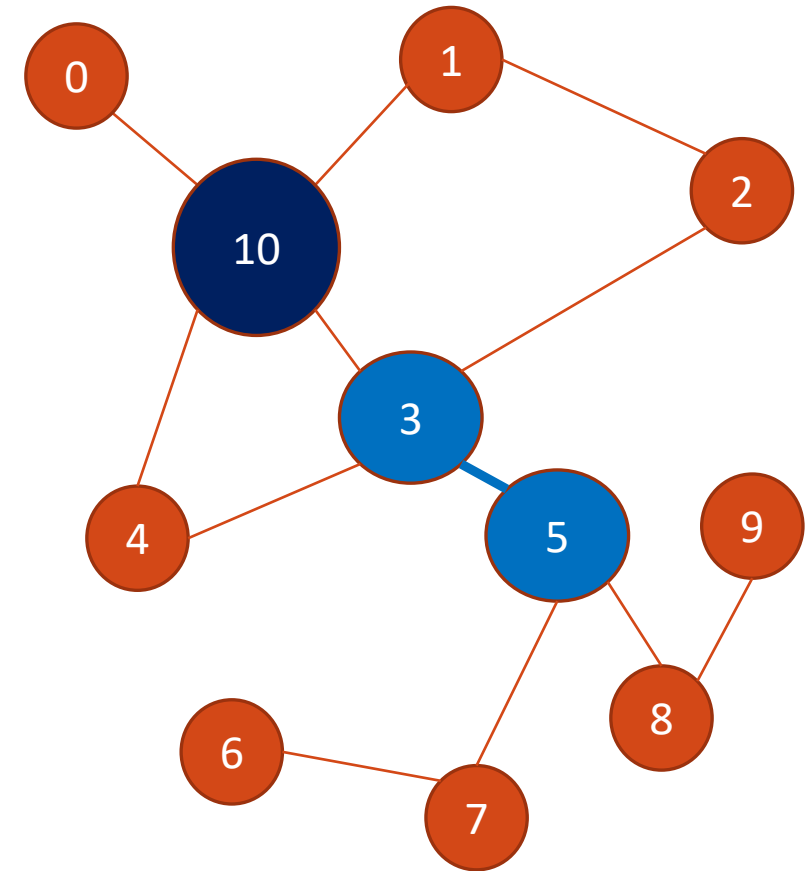
- **Degree** How many people can this person reach directly?
Network of music collaborations: How many people has he/she collaborated with?
- **Betweenness** How likely is this person to be on the most direct route in the network?
Network of spies: Through whom is most of the information likely to flow?
- **Closeness** How fast can this person reach everyone in the network?
Network of sexual relations: How fast will an STD spread to the rest of the network?
- **Eigenvector** How well is this person connected to other well-connected people?
Network of citations: Who is the author cited most by other well-cited authors?

Identifying sets of key players

Example to the right:

- node 10 is the most central according to degree centrality
- ✗ nodes 3 and 5 together will reach more nodes
- ✗ the tie between node 3 and 5 is critical
 - ➔ if severed, the network will break into two isolated sub-networks
- ➔ other things being equal, the nodes 3 and 5 together are more important to this network than node 10 alone

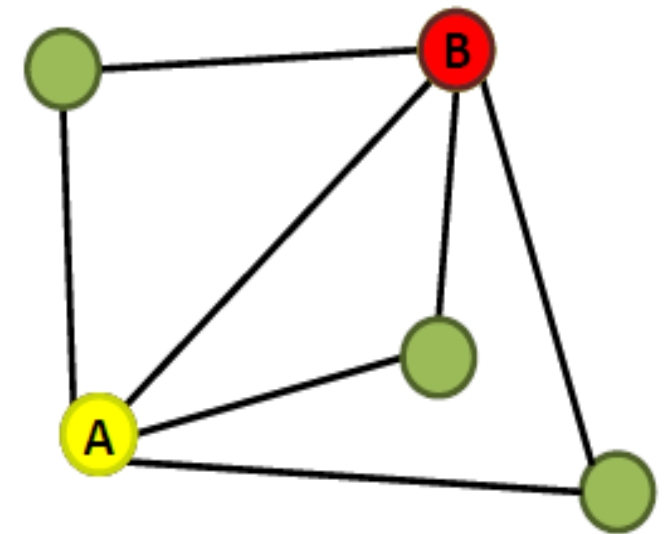
Thinking about sets of key players can help!



Identifying sets of key players

Structural equivalence:

- ~ expresses the similarity between actors in a social network
- is based on the neighbors they share (or, equivalently, the number of identical ties they have)
- **two actors are structurally equivalent if their positions can be swapped without modifying the overall structure of the network**
- **approximate structural equivalence** can be used for hierarchical clustering of social networks

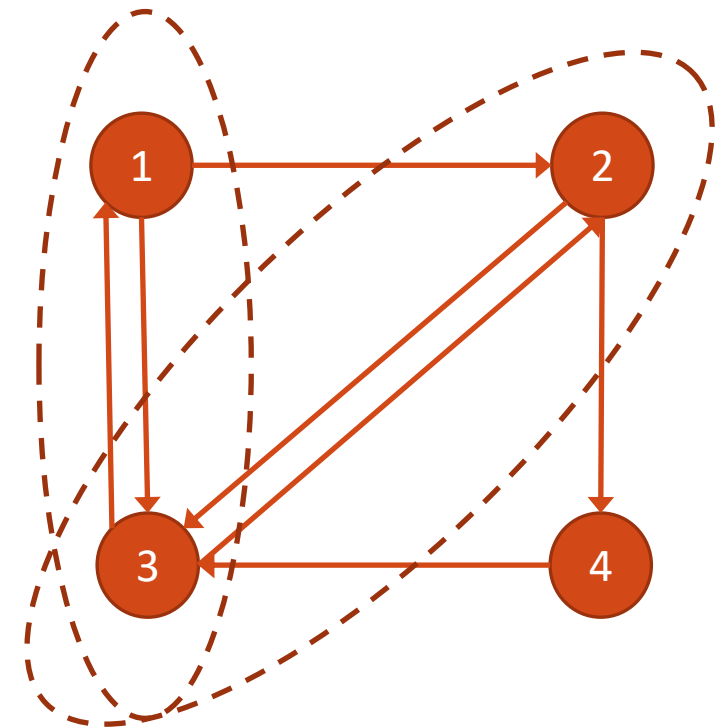


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- **Cohesion** How to characterize a network's structure?

Reciprocity (degree of)

- degree of mutuality and reciprocal exchange in networks, which is **related to social cohesion**
 - ~ ratio of the number of relations that are reciprocated (there is an edge in both directions) over the total number of relations in the network
- ...where two vertices are said to be related if there is at least one edge between them
- makes sense only in directed graphs

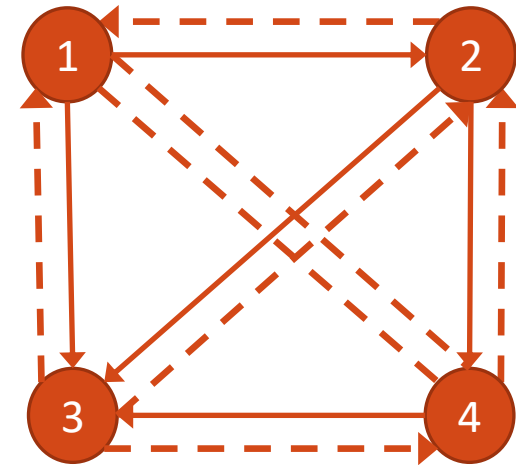
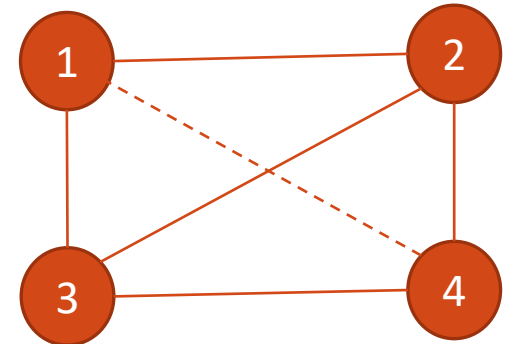


Reciprocity for this network is $2/5 = 0.4$

Density – dense versus sparse networks

- ~ the ratio of the number of edges in the network over the total number of possible edges (that is equal to $\frac{n(n-1)}{2}$, where n is the number of vertices, for an undirected graph)
- a common measure of how well is connected a network (i.e., how closely knit it is)
- a perfectly connected network is called a **clique** and has a density equal to 1
- directed graphs have half the density of their undirected counterparts, as there are twice as many possible edges, i.e., $n(n-1)$
- compares networks (or their parts) against each other

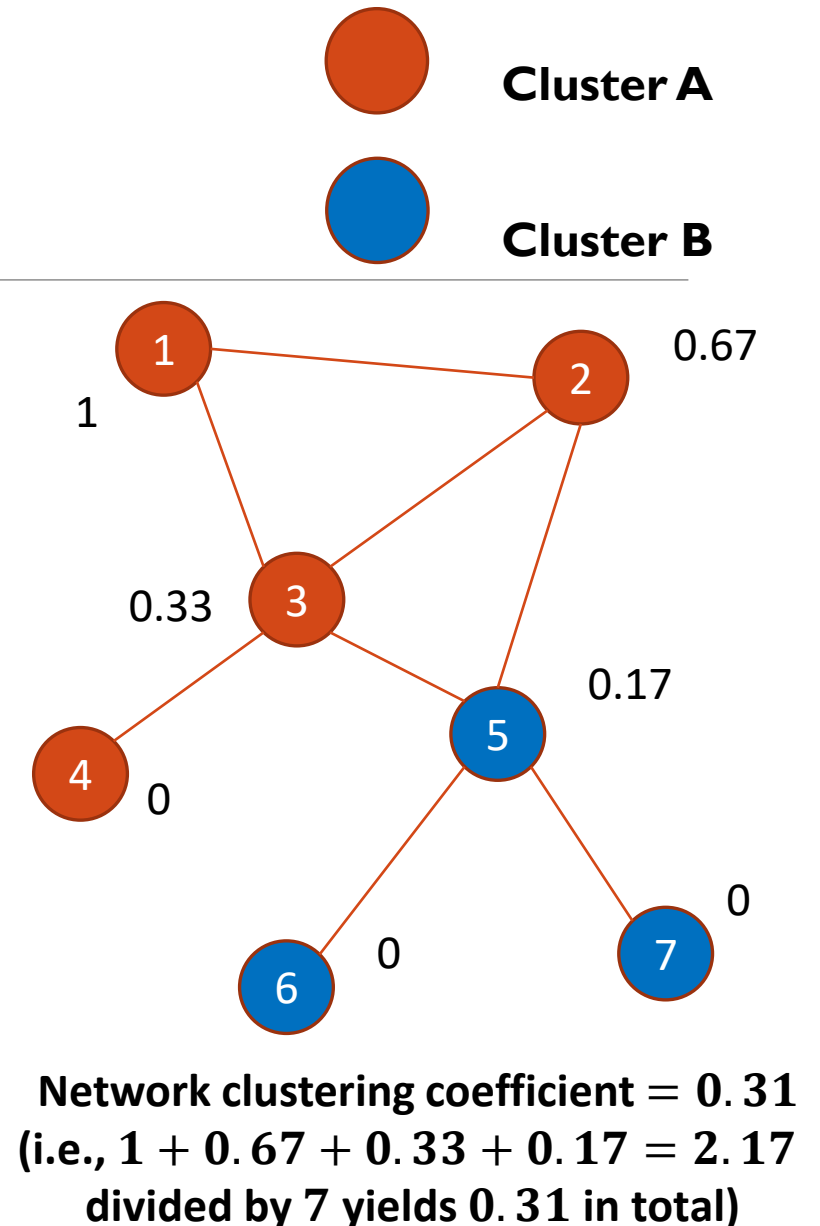
density = $5/6 = 0.83$



density = $5/12 = 0.42$

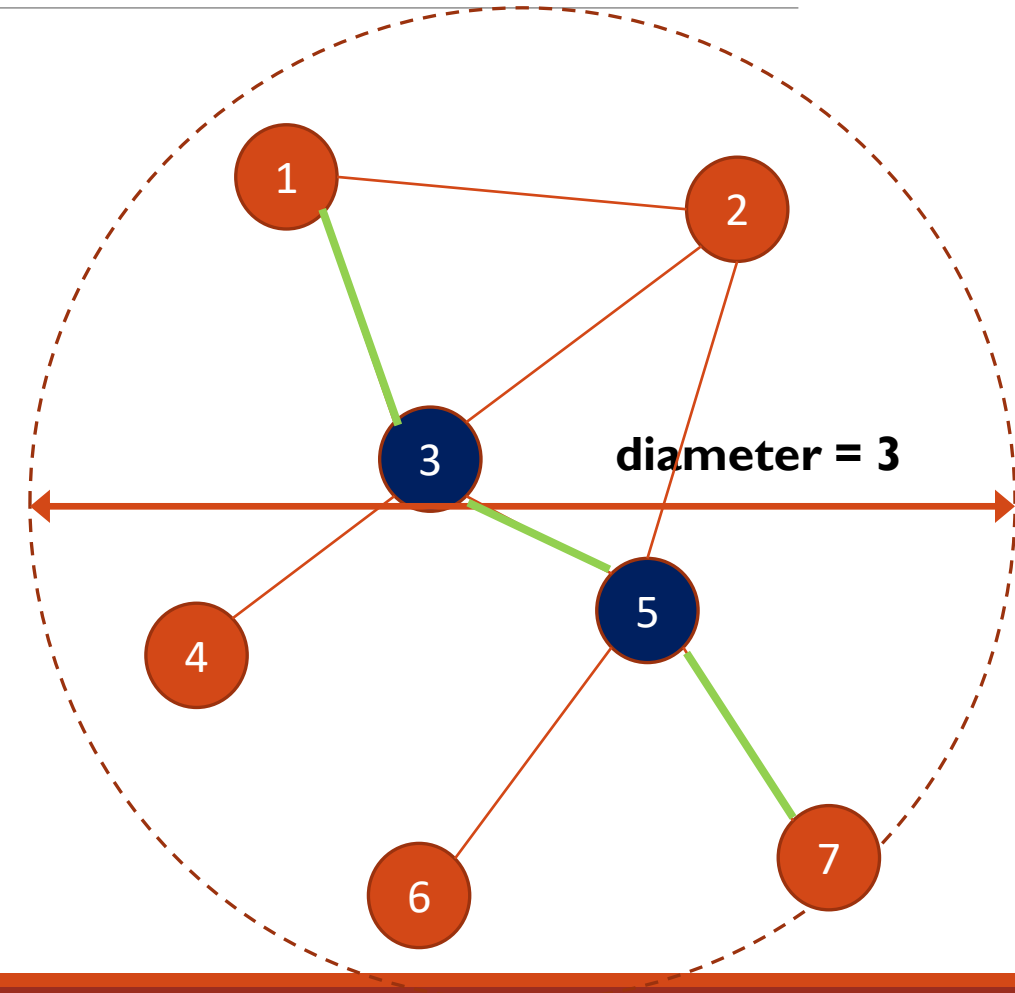
Clustering

- indicates the presence of (sub-)communities
- a **node's clustering coefficient** $C(i)$ measures the density of its neighborhood (i.e., the network consisting only of all the nodes directly connected to it)
 - for a graph $G = (V, E)$, the neighborhood $N(i)$ of a vertex $i \in V$ is defined as $N(i) = \{j \in V; e(i, j) \in E \vee e(j, i) \in E\}$; $k(i) = |N(i)|$
 - $C(i) = \frac{2 |\{e(j, k) \in E; j, k \in N(i)\}|}{k(i) (k(i) - 1)}$ for an undirected graph
 - (or $C(i) = \frac{|\{e(j, k) \in E; j, k \in N(i)\}|}{k(i) (k(i) - 1)}$ for a directed graph, resp.)
 - the **network clustering coefficient** C is the average of all coefficients for its nodes: $C = \sum_{i=1}^{|V|} C(i)$
- **clustering algorithms** try to maximize the number of edges that fall within the same cluster (two clusters identified in the example right)



Average and longest distance

- **network's diameter** ~ the longest shortest path (distance) between any two nodes in a network
- a useful measure of the reach of the network
- indicates how long it will take at most to reach any node in the network
 - sparser networks will generally have greater diameters
- the average of all shortest paths in a network indicates how far apart any two nodes will be on average (average distance)

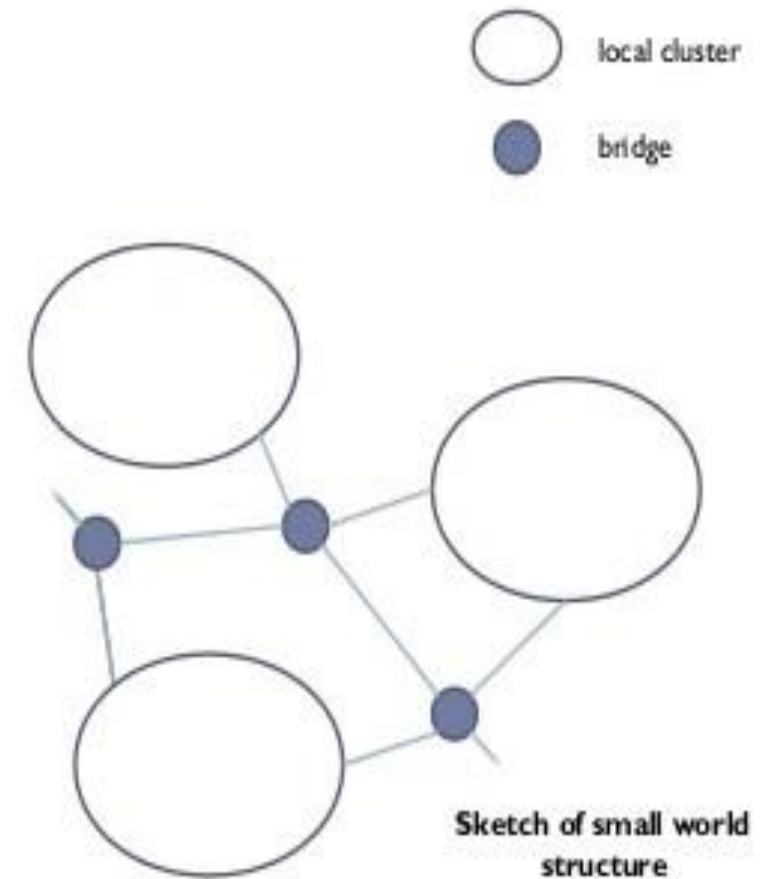


Small Worlds

A **small world** ~ a network that looks almost random but exhibits a significantly **high clustering coefficient** and a relatively **short average path length**

~ **nodes cluster locally and can be reached in a few steps**

- **very common in social networks** because of:
 - transitivity in strong social ties and
 - the ability of weak ties to reach across clusters
- **'small world' networks have many clusters** but also **many bridges** between the clusters that help shorten the average distance between nodes



“6 degrees” of separation are well-known

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Social connections – strong and weak ties

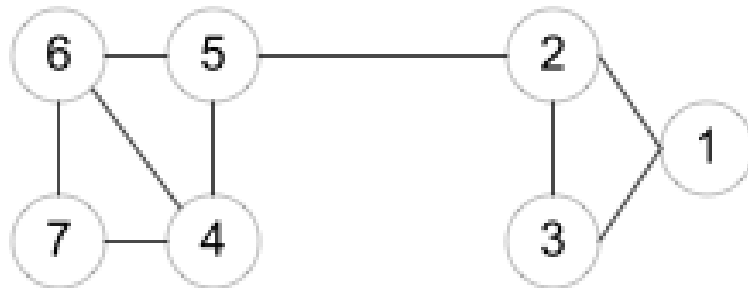
Social media connect users to each other extremely easily

- users might have thousands of friends online
- x but in practice, connections are not of the same strength and can play different roles in community formation and information diffusion
 - **strong ties ~ close friends** and **weak ties ~ acquaintances**
 - **strength of weak ties** (*Granovetter, 1973*) – occasional encounters with distant acquaintances can provide important information about new opportunities for job search
- How to **estimate the strengths of ties** for advanced analysis?
 - analyze network topology
 - learn from user profiles and attributes
 - learn from user activities

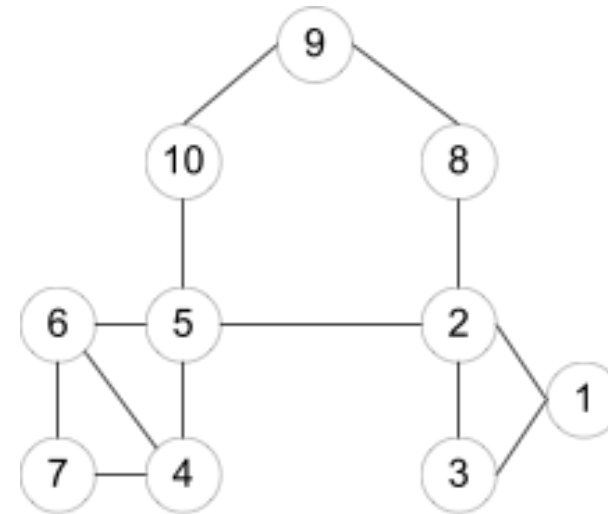
Learning the strength of the ties from the network topology

Bridges connecting two different communities **are weak ties**

- an edge is **a bridge** if its removal results in disconnection of its terminal nodes



$e(2,5)$ is a bridge



$e(2,5)$ is **NOT** a bridge

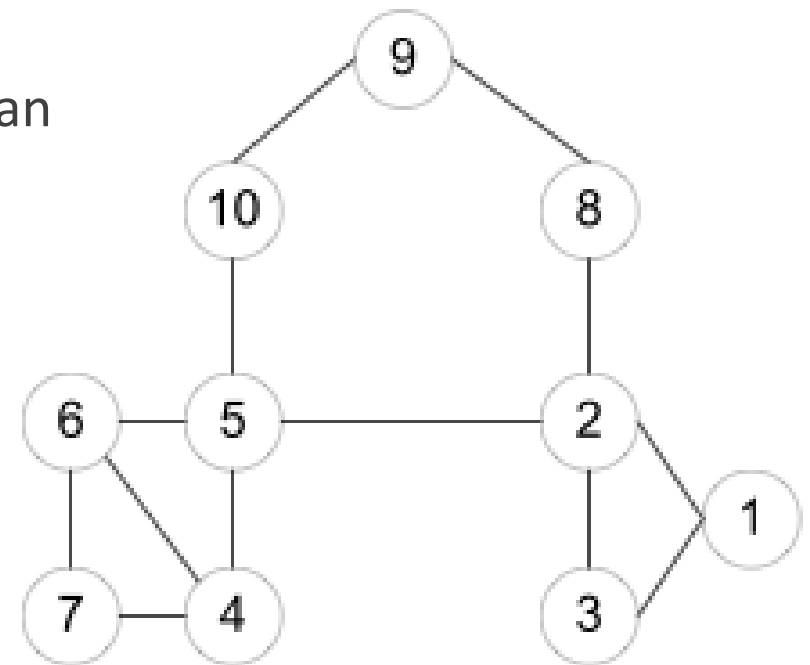
'Shortcut' bridge

Bridges are relatively rare in real-life networks

- relax the definition by checking if **the distance** between two terminal nodes increases (to more than 2) if the edge is removed

=> the larger the distance, the weaker is the tie

- $d(2,5) = 4$ if $e(2,5)$ is removed
- $d(5,6) = 2$ if $e(5,6)$ is removed
- $e(5,6)$ is a stronger tie than $e(2,5)$



Neighborhood overlap for two nodes connected by a tie

The strength of the ties can be measured based on neighborhood overlap

→ the larger the overlap, the stronger is the tie (between v_i and v_j)

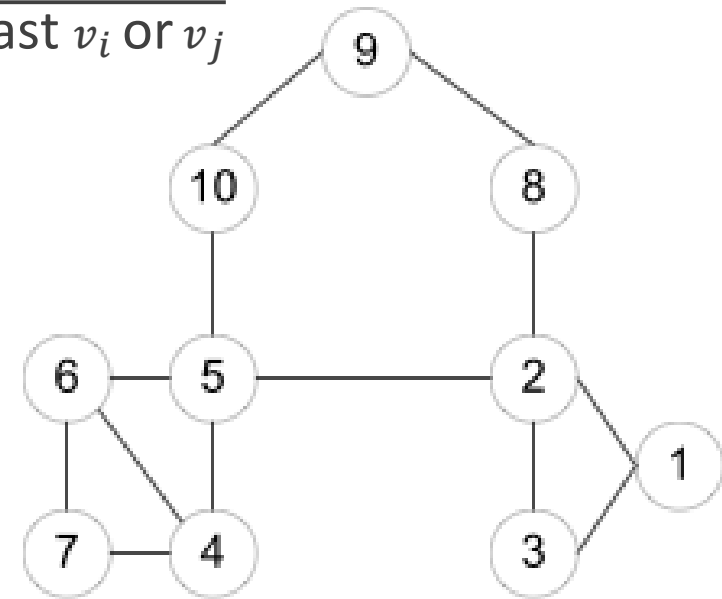
$$\text{overlap}(v_i, v_j) = \frac{\text{number of shared friends of both } v_i \text{ and } v_j}{\text{number of friends who are adjacent to at least } v_i \text{ or } v_j}$$

$$= \frac{|N_i \cap N_j|}{|N_i \cup N_j| - 2} \quad (-2 \text{ in the denominator is to exclude } v_i \text{ and } v_j)$$

$$\text{overlap}(2, 5) = 0$$

$$\text{overlap}(5, 6) = \frac{|\{4\}|}{|\{2, 4, 5, 6, 7, 10\}| - 2} = 1/4$$

for the graph $G = (\{1, 2\}, \{(1, 2)\})$, $\text{overlap}(1, 2) =_{df} 0$



Learning the strength of the ties from profiles and interactions

Twitter (one can follow others without their confirmation)

- the real friendship network is determined by the frequency with which two users talk to each other, rather than the follower-followee network
- the real friendship network is more influential in driving Twitter usage

Facebook: strengths of ties can be predicted based on various information, e.g.,

- friend-initiated posts, messages exchanged in wall posts, number of mutual friends, etc.

Learning **numeric** link strength, e.g., by maximum likelihood estimation

- user profile similarity determines the strength
 - link strength, in turn, determines user interaction
- ➔ **maximize the likelihood based on observed profiles and interactions**

Learning the strength of the ties from user activities

- We can learn how the users influence their friends if the user activity log is accessible
- Maximize the likelihood of user activity depending on the adopted influence modeling method
 - Linear threshold model (LTM)
 - Independent cascading model (ICM)
- **Influence modeling** is important **for understanding information diffusion, spread of new ideas, and word-of-mouth (viral) marketing**

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Influence Modeling

- Linear Threshold Model
- Independent Cascade Model
- Influence vs. Correlation